

Coding Challenge 4_Markdown

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1 Question 1

****YAML** The YAML header is the section at the top of the R Markdown file enclosed by `---`. It contains data about title, author name, date of markdown creation, output format and table of contents.

****Literate Programming** Literate programming is a style where you write code and explanations together in the same document.

This helps readers understand why we wrote the code, not just what the code does.

2 Question2a

-[PaperLink]<https://doi.org/10.1094/PDIS-06-21-1253-RE>

3 Question2b

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.2
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
## Warning: package 'tibble' was built under R version 4.5.2
```

```
## Warning: package 'tidyr' was built under R version 4.5.2
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```
## Warning: package 'purrr' was built under R version 4.5.2
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
## Warning: package 'stringr' was built under R version 4.5.2
```

```
## Warning: package 'forcats' was built under R version 4.5.2
```

```
## Warning: package 'lubridate' was built under R version 4.5.2
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.6
```

```
## v forcats   1.0.1      v stringr   1.6.0
```

```
## v ggplot2   4.0.2      v tibble    3.3.1
```

```
## v lubridate 1.9.4      v tidyr     1.3.2
```

```
## v purrr     1.2.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

```
library(ggpubr)
```

```
## Warning: package 'ggpubr' was built under R version 4.5.2
```

```
mycotoxin<- read.csv("MycotoxinData.csv", na.strings="na")
```

```
mycotoxin$Treatment <- as.factor(mycotoxin$Treatment)
```

```
mycotoxin$Cultivar <- as.factor(mycotoxin$Cultivar)
```

```
mycotoxin$DON <- as.numeric(mycotoxin$DON)
```

```
mycotoxin$X15ADON <- as.numeric(mycotoxin$X15ADON)
```

```
mycotoxin$MassperSeed_mg <- as.numeric(mycotoxin$MassperSeed_mg)
```

```
str(mycotoxin)
```

```
## 'data.frame': 375 obs. of 6 variables:
## $ Treatment : Factor w/ 5 levels "Fg","Fg + 37",...: 1 1 1 1 1 1 1 1 1 ...
## $ Cultivar : Factor w/ 2 levels "Ambassador","Wheaton": 2 2 2 2 2 2 2 2 2 ...
## $ BioRep : int 2 2 2 2 2 2 2 2 2 3 ...
## $ MassperSeed_mg: num 10.29 12.8 2.85 6.5 10.18 ...
## $ DON : num 107.3 32.6 416 211.9 124 ...
## $ X15ADON : num 3 0.85 3.5 3.1 4.8 3.3 6.9 2.9 2.1 0.71 ...
```

4 Question2c

```
cbbPalette <- c("#56B4E9", "#009E73")
```

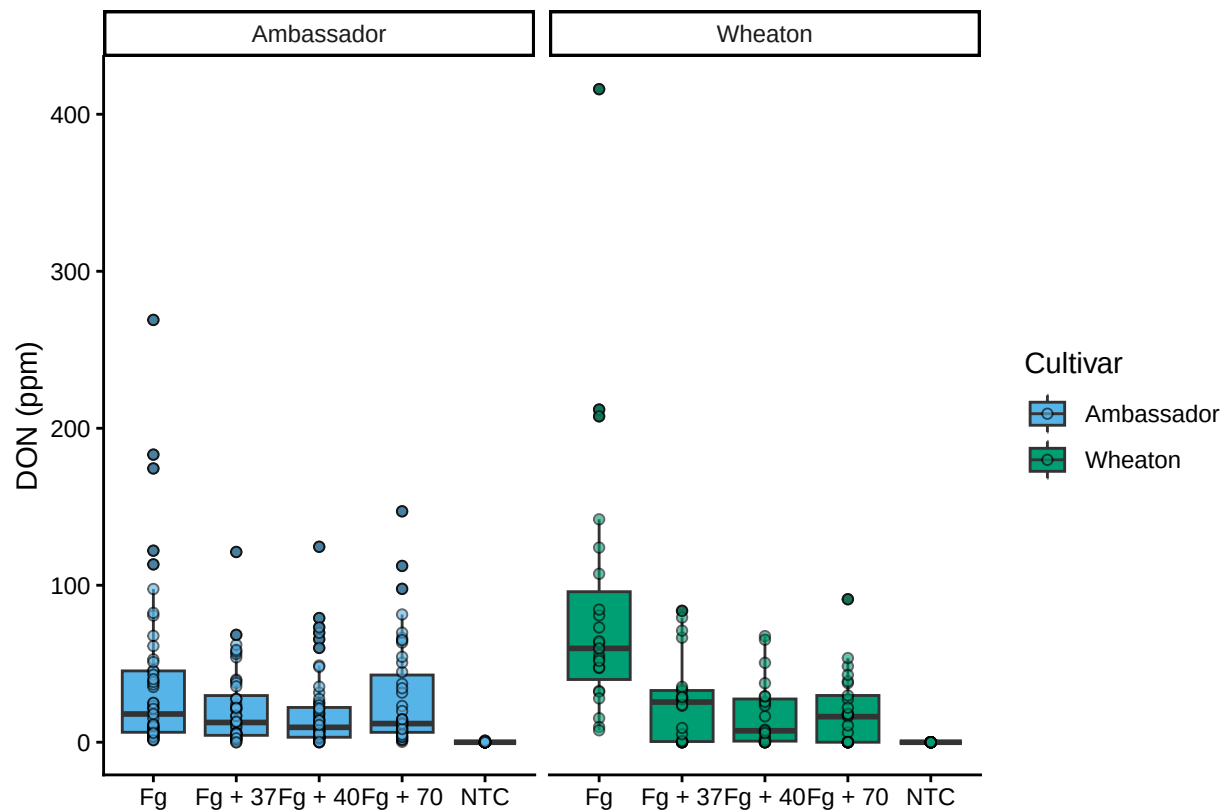
5 DON_PLOT1

```
Plot1 <- ggplot(mycotoxin, aes(x = Treatment, y = DON, fill = Cultivar)) +
  geom_boxplot(position = position_dodge(0.85)) +
  xlab("") +
  ylab("DON (ppm)") +
  geom_point(alpha = 0.6, pch = 21, color = "black") +
  scale_fill_manual(values = cbbPalette) +
  facet_wrap(~Cultivar) +
  theme_classic()
```

Plot1

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range
## ('geom_point()').
```

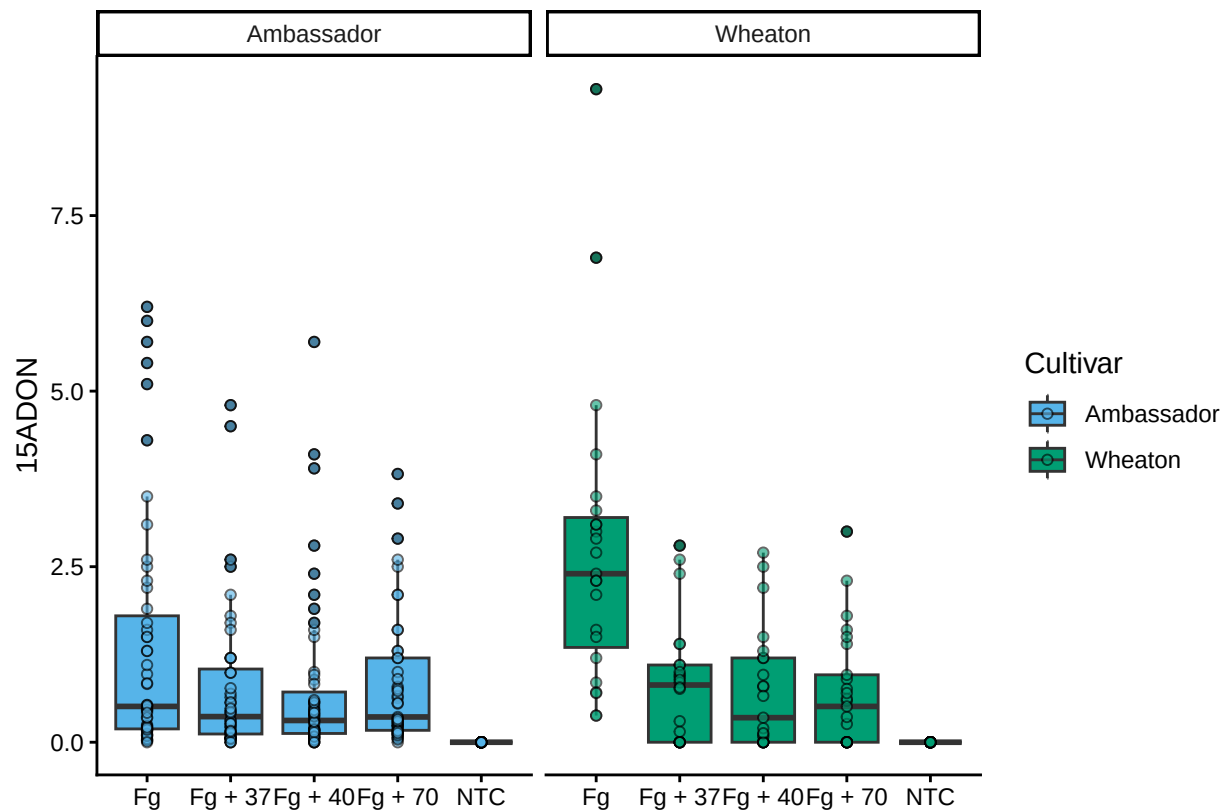


6 Plot X15ADON

```
Plot2 <- ggplot(mycotoxin, aes(x = Treatment, y = X15ADON, fill = Cultivar)) +
  geom_boxplot(position = position_dodge(0.85)) +
  xlab("") +
  ylab("X15ADON") +
  geom_point(alpha = 0.6, pch = 21, color = "black") +
  scale_fill_manual(values = cbbPalette) +
  facet_wrap(~Cultivar) +
  theme_classic()
Plot2
```

```
## Warning: Removed 10 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_point()').
```



7 SeedMass

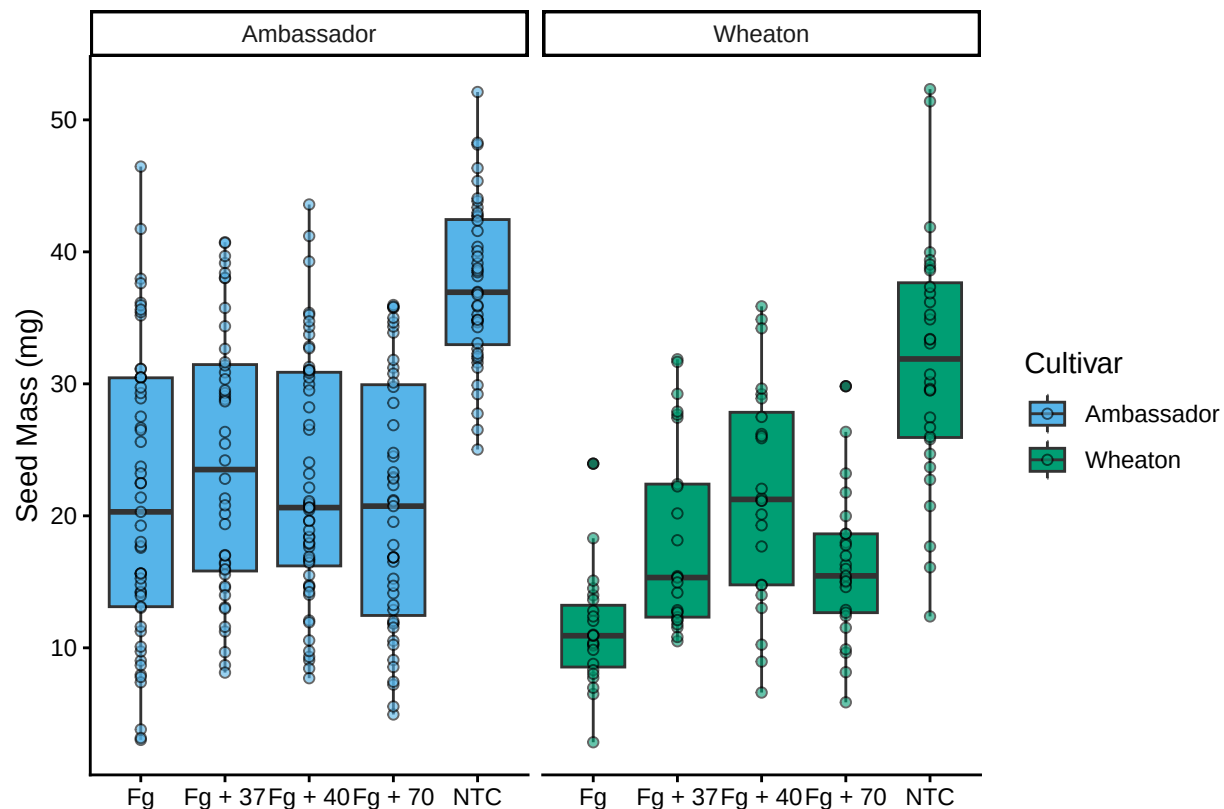
#Plot 3

```
Plot3<- ggplot(mycotoxin, aes(x = Treatment, y = MassperSeed_mg, fill = Cultivar)) +
  geom_boxplot(position = position_dodge(0.85)) +
  xlab("") +
  ylab("Seed Mass (mg)") +
  geom_point(alpha = 0.6, pch = 21, color = "black") +
  scale_fill_manual(values = cbbPalette)+
  facet_wrap(~Cultivar)+
  theme_classic()
```

Plot3

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
#3Combined_PLOT
```

```
library(ggpubr)
```

```
figure=ggarrange(Plot1,Plot2,Plot3,labels = c("A", "B", "C"),nrow = 1,ncol = 3, common.legend=TRUE)
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range
## ('geom_point()').
```

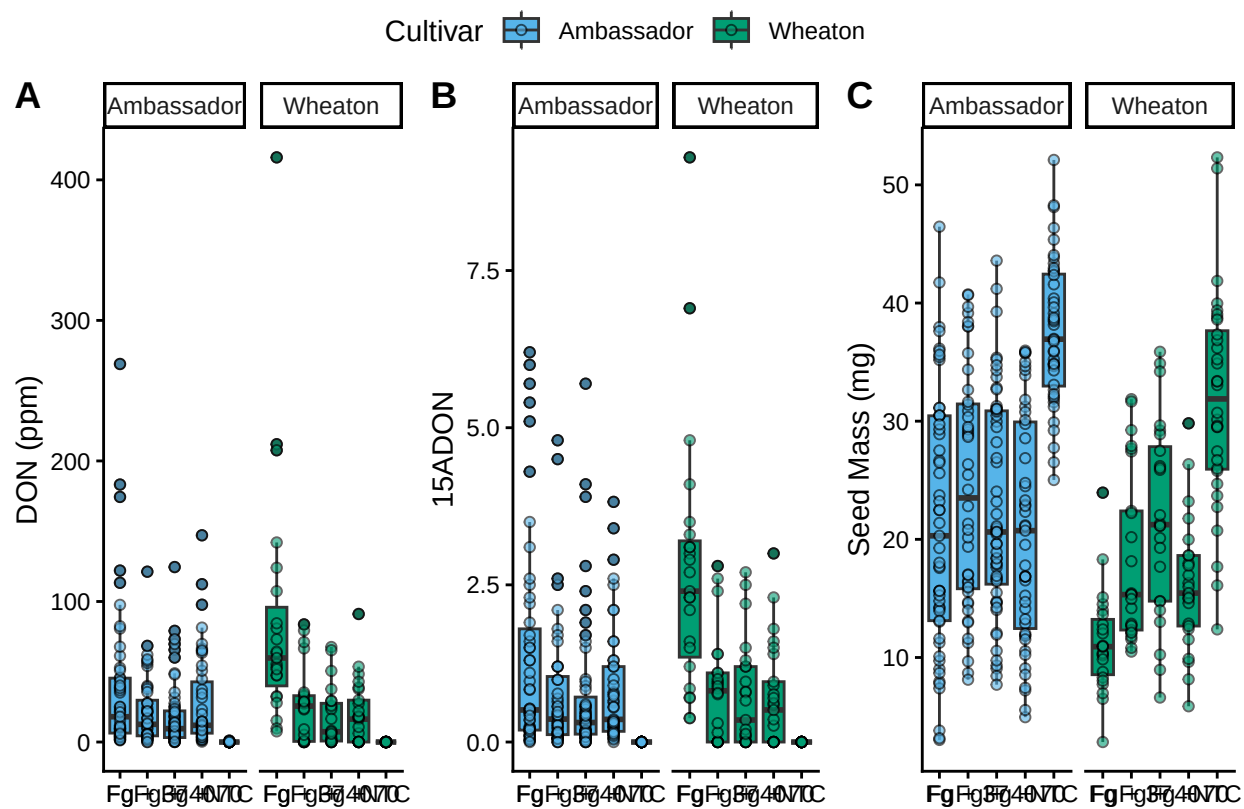
```
## Warning: Removed 10 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```

figure



```
#Combined_with_pairwisecomparison
```

```
## Plot 1: plot1 with pairwise comparisons
```

```
Plot1_pwc <- Plot1 +
```

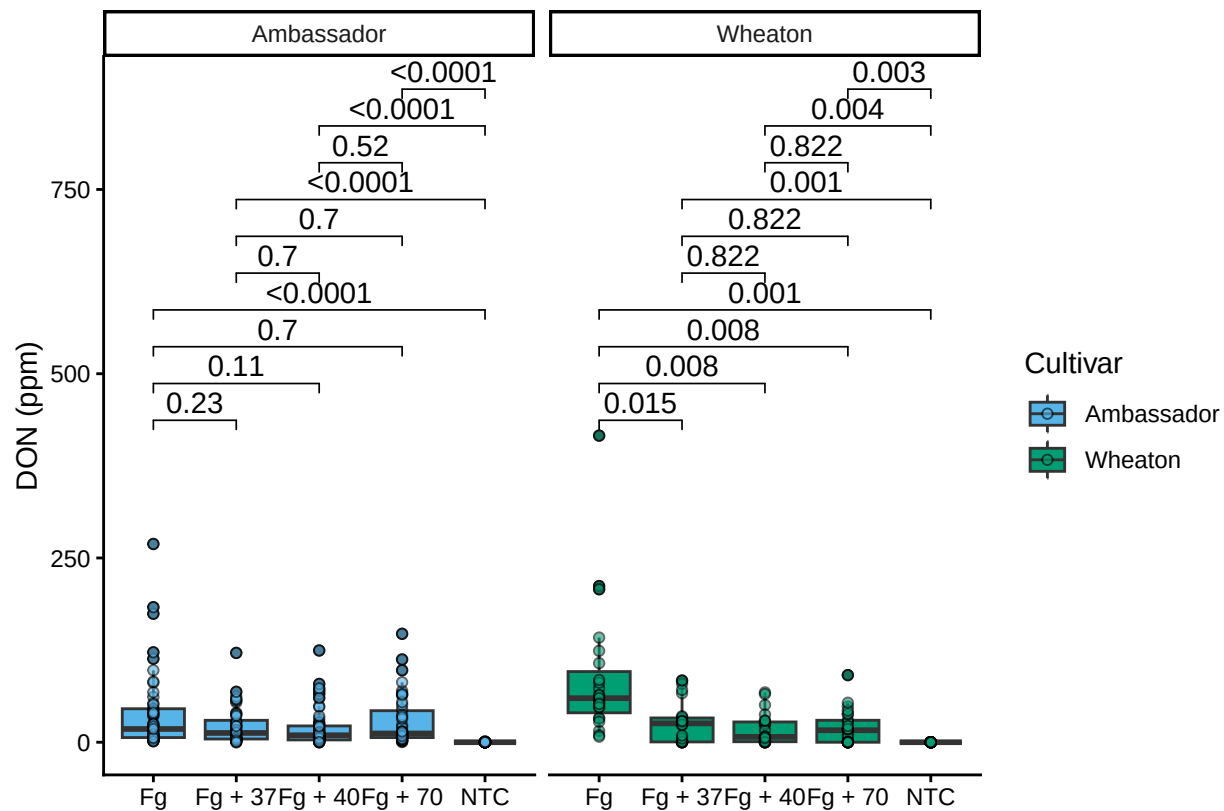
```
  geom_pwc(aes(group = Treatment), method = "t.test", label = "p.adj.format") + # Add pairwise t-tests
  theme_classic()
```

```
Plot1_pwc
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## ('stat_pwc()').
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
##Plot 2: plot2 with pairwise comparisons
```

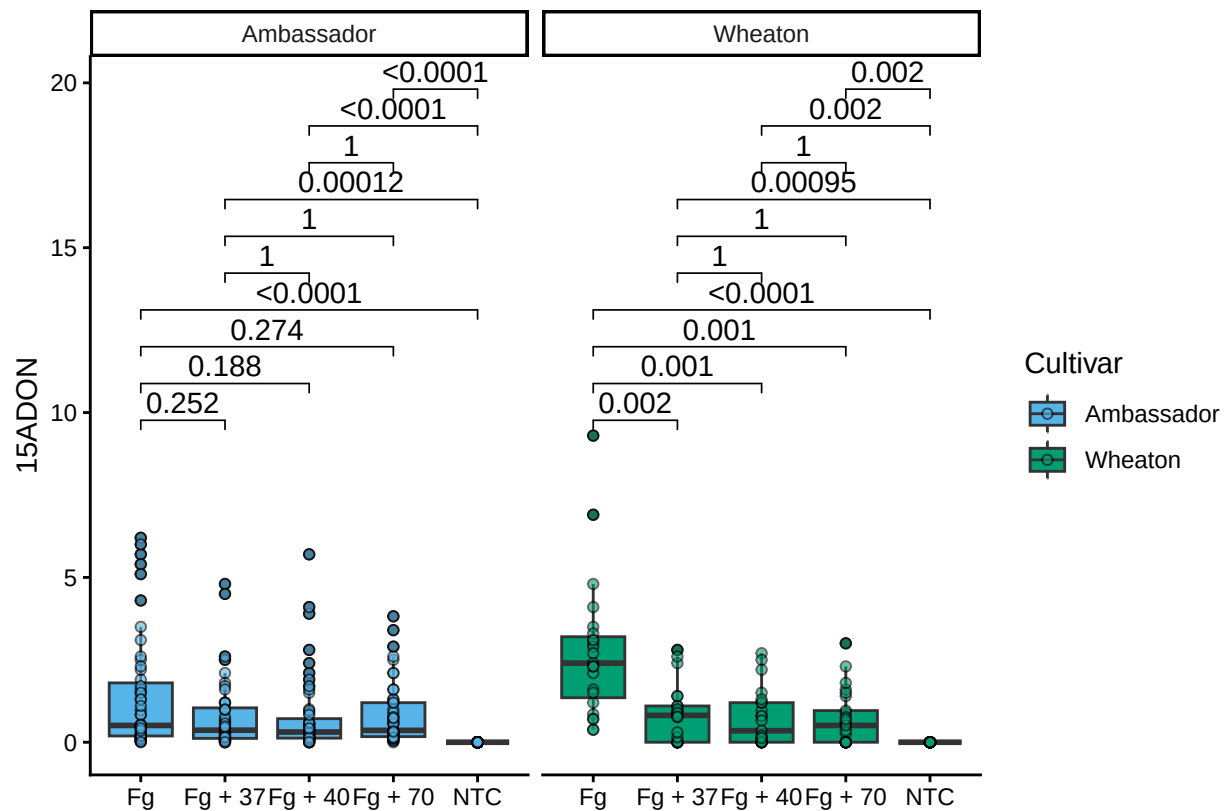
```
Plot2_pwc <- Plot2 +
  geom_pwc(aes(group=Treatment), method= "t.test", label= "p.adj.format")+
  theme_classic()
```

```
Plot2_pwc
```

```
## Warning: Removed 10 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 10 rows containing non-finite outside the scale range
## ('stat_pwc()').
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
##Plot 3: plot3 with pairwise comparisons
```

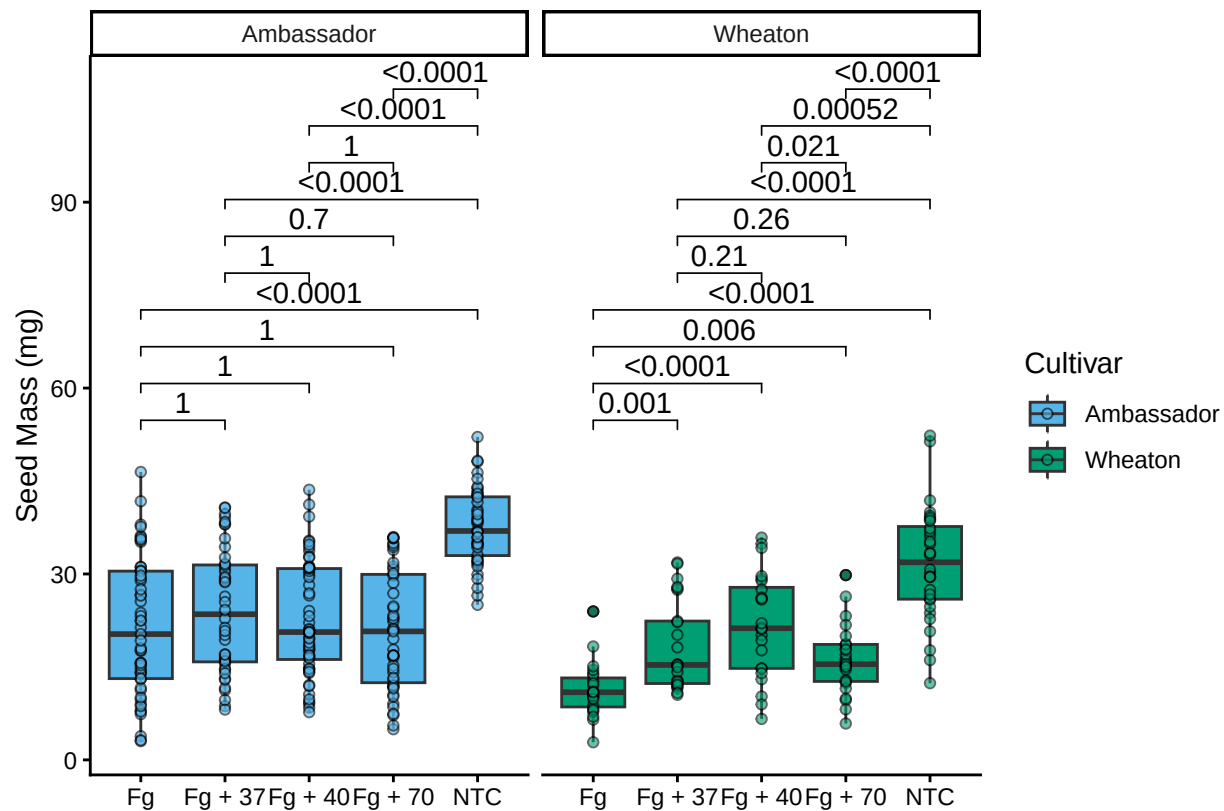
```
Plot3_pwc = Plot3 + geom_pwc(aes(group=Treatment), method="t.test", label="p.adj.format")+  
  theme_classic()
```

```
Plot3_pwc
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range  
## ('stat_boxplot()').
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range  
## ('stat_pwc()').
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



#Combine all three plots with a common legend ----

```
Combined_pwc = ggarrange(Plot1_pwc, Plot2_pwc, Plot3_pwc, labels = c("D", "E", "F"), nrow = 1, ncol = 3, common.legend = TRUE)
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## ('stat_pwc()').
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## ('stat_pwc()').
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 10 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 10 rows containing non-finite outside the scale range
## ('stat_pwc()').

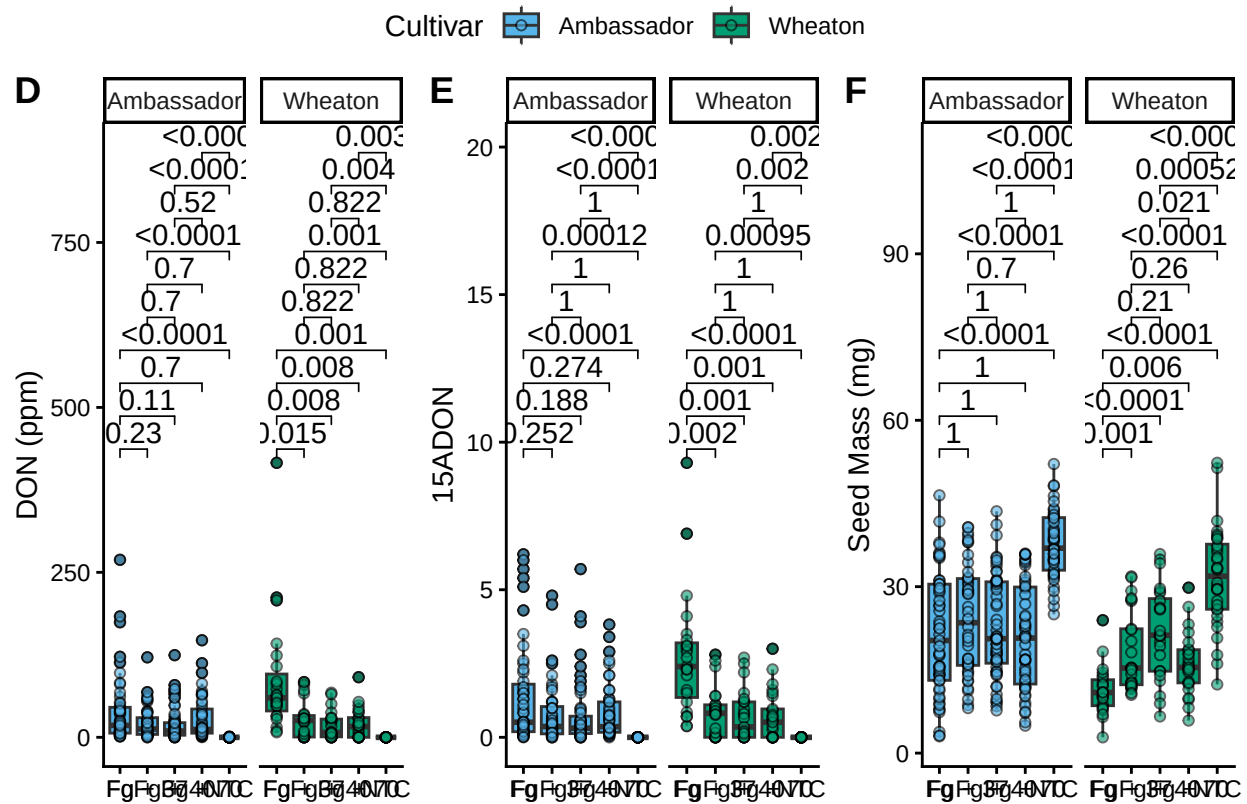
## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_point()').

## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_boxplot()').

## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_pwc()').

## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Combined_pwc



8 Question5

README- <https://github.com/Hikmat-au/Assignments-repository/blob/main/Coding%20Challenge4/Coding-challenge-4.md>

9 Question6

Github- [https://github.com/Hikmat-au/Assignments-repository/tree/main/Coding%20Challenge4\(\)](https://github.com/Hikmat-au/Assignments-repository/tree/main/Coding%20Challenge4())